

EarComp 2021: Second International Workshop on Earable Computing

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ABSTRACT

The objective of the 2nd ACM International Workshop on Earable Computing (EarComp 2021) is to provide an academic forum and bring together researchers, practitioners, and design experts to discuss how sensory earables technologies have and can complement human sensing research. It also aims to provide a launchpad for bold and visionary ideas and serve as a catalyst for advancements in this emerging new Earable Computing research space.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile devices; Ubiquitous and mobile devices; • Computing methodologies** → **Machine learning algorithms.**

KEYWORDS

Earable; Human Sensing; Behaviour Analysis; Data Systems

ACM Reference Format:

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1 BACKGROUND

Sensory earables are increasingly becoming a mainstream computing platform with a promise to fundamentally transform personal-scale human sensing applications. Over the past few years, a number of research efforts in the ubiquitous computing domain have sought to achieve useful, engaging, and sometimes ambitious behavioural analytics with sensory earables including studies of the human

face; of emotion and stress; continuous monitoring of cardiovascular function; oxygen consumption and blood flow; and tracking eating episodes as well as dietary and swallowing activities. At the same time, we have started seeing commercial efforts such as Apple AirPods, Bragi's The Dash, Bose SoundSport, Jabra Elite Sport, and Sony Xperia offering music experience augmented with sensory services including fitness tracking, real-time translations and conversational agents. Naturally, earables are becoming an intense interdisciplinary area of study with many and diverse applications including HCI, empathic communication, behavioural science, health and wellbeing, entertainment, education, and security.

However, as of today, earable computing lacks an academic forum to bring together researchers, practitioners, and design experts from academia and industry to discuss, share, and shape this exciting new area of research. To this end, we want to organise the second edition of the ACM Workshop on Earable Computing with a hope that this workshop will serve as a catalyst for advancements in sensory earable technology as well as present a clear sense of direction for the research community to proceed in this space.

As a launchpad, we leveraged the Open Earable Platform, eSense (<http://www.esense.io>), from Nokia Bell Labs. These devices are being shared with 70+ academic institutions to accelerate the research in this space. In particular, we expect the workshop will include participants from these institutions reporting on their research with sensory earables to bootstrap this community. For the second edition of the workshop, we will also consider research contributions which do not build on top of eSense but present innovative developments with other earable devices.

2 ESENSE AND EARABLE COMPUTING COMMUNITY

Nokia Bell Labs Cambridge developed an open earable computing platform called eSense in 2019. eSense provides a high-definition stereo earable equipped with a microphone, a 6-axis inertial measurement unit, and dual-mode Bluetooth. It allows recording of real-time data streams of three modalities: audio, motion, and proximity. The platform (hardware and associated software libraries) is being widely shared with 70+ academic groups active in the areas of ubicomp, mobile and wearable systems research. It is expected that these groups will contribute to building a community around earable computing by sharing:

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- curated and labelled datasets collected with eSense across a variety of projects; and
- models using these datasets.

The eSense portal (<http://www.esense.io>) will be the central point of access for this community, and will host project descriptions, links to data and models, relevant publications of different research groups as well as the EarComp workshop page. We hope, through this portal, academic research groups working on this topic will develop an active community space.

3 SCOPE, SUBMISSIONS, STRUCTURE AND SESSIONS

3.1 Call for Papers

We will solicit two categories of papers. Full papers (up to 6 pages including references) should report a reasonably mature work with earables, and is expected to demonstrate concrete and reproducible results albeit scale may be limited. Interactive poster papers (up to 2 pages including references) are encouraged to report novel, and creative ideas that are yet to produce concrete research results but are at a stage where community feedback would be useful. Moreover, we will have a special submission category - "Dataset Paper" - soliciting a 1-2 page document describing a well curated and labelled dataset collected with earables (eventually accompanied by the dataset). Full research papers will be in ACM sigconf template with 2 columns and the accepted papers will be included in the ACM Digital Library. All papers will be digitally available through the workshop website, and the UbiComp adjunct proceedings. For each category of papers, we will offer a "Best Paper" and "Best Dataset" awards sponsored by Nokia Bell Labs. In addition, depending on the quality and depth of the submissions we might consider producing a Book on "Earable Computing" contributed by the authors of the papers, and edited by the Workshop Organisers.

Topics of interest:

- Acoustic Sensing with Earables
- Kinetic Sensing with Earables
- Multi-Modal Learning with Earables
- Multi-Task Learning with Earables
- Active Learning with Earables
- Low-Power Sensing Systems for Earables
- Authentication & Trust mechanisms for Earables
- Quality-Aware Data Collection with Earables
- Experience Sampling with Earables
- Crowd Sourcing with Earables
- Novel UI and UX for Earables
- Auditory Augmented Reality Application with Earables
- Health and Wellbeing Applications of Earables
- Emerging applications of Earables

3.2 Workshop Program

In the second edition of the workshop, we accepted nine contributions from academic research groups, covering a broad area of research topic related to earable computing.

The 1-day workshop programme is intended to be an interactive one and will include a plenary keynote, several technical sessions, a focused panel session. We plan to invite one of the luminaries in

the area of mobile and wearable systems research as the keynote speaker to set the tone of the workshop. This will be followed by technical sessions with nine papers, each focusing on specific themes related to Earable Computing:

- (1) Systems for Earables
- (2) Learning Algorithms for Earables
- (3) Novel Lifestyle Applications with Earables

Each session will host several long talks (15 minutes). We will close the workshop by organising a 4-member panel session in which we will bring together some of the key researchers from the academia and industry to discuss the grand challenges for earables shaped by the lessons from their success (and failures) with computing on small devices. The panel will also strive to define 2-3 key challenges that the earable computing community at large should address before the next edition of the workshop.

We intend to keep the workshop open for all UbiComp 2021 attendees. However, we might need to limit attendees depending on the number of submitted research papers.

3.3 Organisation

- Website: Earable Computing Workshop will be hosted at <https://www.esense.io/earcomp2021/index.html>
- Publicity: The workshop will already have visibility through the eSense portal, and the participating research groups. In addition, we will maintain active community engagement through twitter and mailing lists.
- Paper Management: A PCS portal will be setup to manage the submissions.

4 EARCOMP 2019

The current proposal is towards organising the 2nd edition of the EarComp workshop and looks forward to holding its tradition of being co-located with UbiComp/ISWC. In the first edition of the workshop, we accepted ten contributions from academic research groups that are working with the eSense computing platform. These contributions covered a broad area of research topics related to earable computing including kinetic sensing, quality-aware data collection and machine learning, data platforms and auditory augmented reality, as well as health and wellbeing applications. The workshop included technical papers from leading researchers in the field working on earables including KIT, Georgia Tech, Cornell University, University of Cambridge, Google Brain, Apple, and Samsung. The workshop also had a keynote talk from Prof. Romit Roy Choudhury (University of Illinois Urbana-Champaign) and featured a panel session with a focus on ubiquitous sensing with earables in which leading researchers in the field debated the future directions of earables and their interaction with other sensor systems. The panellists were Prof. Cecilia Mascolo (University of Cambridge), Prof. Michael Beigl (Karlsruhe Institute of Technology) and Dr. Thomas Ploetz (Georgia Institute of Technology). Figure 1 shows the photos of EarComp 2019. More photos can be found at the [link](#).



Figure 1: EarComp 2019

5 PROGRAM COMMITTEE MEMBERS

The program committee members comprise a set of eminent experts in diverse fields relating to earable systems. The names of the program committee members are as follows:

- Michael Beigl, Karlsruhe Institute of Technology
- Marios Costantinides, Nokia Bell Labs
- Nigel Davies, Lancaster University
- Inseok Hwang, Pohang University of Science and Technology
- Seungwoo Kang, Korea University of Technology and Education
- JeongGil Ko, Ajou University
- Nicholas Lane, University of Cambridge
- Youngki Lee, Seoul National University
- Dong Ma, University of Cambridge
- Cecilia Mascolo, University of Cambridge
- Tatsuo Nakajima, Waseda University
- Tadashi Okoshi, Keio University
- Shijia Pan, UC Merced
- Johannes Schöning, University of Bremen
- June-hwa Song, KAIST
- Mani Srivastava, UCLA
- Kristof Van Laerhoven, University of Siegen
- Moustafa Youssef, Alexandria University

6 WORKSHOP ORGANISERS

Fahim Kawsar is the Founding Director of Pervasive Systems research at Nokia Bell Labs, Cambridge, UK and holds a Design

United Professorship at TU Delft, Netherlands. An experimental computer scientist at heart, he spends most of his time at Cambridge building sensory AI systems with novel computational forms to explain human behaviour with other people, places, and things. He is intrigued by the power of design and frequently visits TU Delft to teach and debate the influence of sensory data systems in redefining design, and vice versa. Fahim together with his Bell Labs team has developed the eSense earables and actively manages the community around it. His work and publications can be viewed at <http://www.fahim-kawsar.net>.

Robert Harle is a Reader (Associate Professor) at the University of Cambridge in the UK and a Fellow in Computer Science at Downing College. His research interests are related to all forms of digital sensing with mobile and wearable systems. He has a particular interest in human movement at all granularities, from geolocation to limb trajectories for technique assessment. He has published in the fields of indoor positioning, context-awareness and ubiquitous computing. Since 2017 he has also been a Visiting Researcher in the Google Wear & Fit team.

Alessandro Montanari is a Research Scientist at Nokia Bell Labs in the Pervasive Systems Research team in Cambridge (UK) and he holds a PhD in Computer Science from the University of Cambridge. At Bell Labs he is part of the team developing the eSense earables and responsible for their distribution to 50+ research institutions. His research interests include expanding and improving the capabilities of modern sensory systems for efficient human sensing. He

is particularly interested in ultra-wearable and ultra-low-power devices with a focus on novel sensing and computational techniques.

Chulhong Min is a Research Scientist at Nokia Bell Labs, Cambridge, UK and a visiting fellow at Department of Computer Science and Technology, University of Cambridge, UK. He received the Ph.D.

degree in Computer Science from KAIST in 2016. He is an Associated Editor of ACM Proceedings on Interactive, Mobile, Wearable and Ubiquitous Technologies. His research interests include mobile and wearable systems, machine learning, the Internet of Things, and human behaviour modelling.